ICE-ID: A Novel Historical Census Dataset for Longitudinal Identity Resolution

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Abstract

We introduce **ICE-ID**, a benchmark dataset comprising 984,028 records from 16 Icelandic census waves spanning 220 years (1703–1920), with 226,864 expert-curated person identifiers. ICE-ID combines hierarchical geography (farm→parish→district→county), patronymic naming conventions, sparse kinship links (partner, father, mother), and multi-decadal temporal drift—challenges not captured by standard product-matching or citation datasets. This paper presents an artifact-backed analysis of temporal coverage, missingness, identifier ambiguity, candidate-generation efficiency, and cluster distributions, and situates ICE-ID against classical ER benchmarks (Abt–Buy, Amazon–Google, DBLP–ACM, DBLP–Scholar, Walmart–Amazon, iTunes–Amazon, Beer, Fodors–Zagats). We also define a deployment-faithful temporal OOD protocol and release the dataset, splits, regeneration scripts, analysis artifacts, and a dashboard for interactive exploration. Baseline model comparisons and end-to-end ER results are reported in the companion methods paper.

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Preprint. Under review.

1 Introduction

Linking historical census records is fundamental to research on social mobility, demographic change, migration, and epidemiology, yet it remains arduous because names mutate, fields are missing, and administrative borders shift over time [23, 1]. Most census-specific benchmarks oversimplify these challenges: they cover only short time ranges (often a single decade), omit kinship structure, and treat geography as flat text rather than a hierarchy [21].

Carefully curated benchmarks can transform entire fields: *ClimSim* unlocked hybrid physics–ML climate modelling [28]; *DecodingTrust* exposed safety gaps in frontier LLMs [26]; the *PRISM Alignment Dataset* broadened evaluation of alignment techniques across diverse regions [8]. Inspired by these successes, we release **ICE-ID**, the first large-scale open benchmark focused on *long-term* person matching in a national population.

This paper focuses on the dataset itself: provenance, schema, statistical properties, and benchmark design for longitudinal identity resolution. Model evaluations and method comparisons are reported in the companion methods paper.

Contributions. We provide:

- a longitudinal census dataset with stable identity labels, hierarchical geography, and optional kinship links;
- a temporal OOD evaluation protocol and task definitions suitable for both pairwise and clustering tracks;
- a set of dataset-centric diagnostics (temporal coverage, missingness, ambiguity, blocking efficiency, cluster-size CCDF) generated from published artifacts;
- a reproducible release with explicit provenance, licensing, a dashboard, and scripts that regenerate all tables and figures in this paper.

2 Related Work

Historical census linkage and record linkage datasets. Linking historical census records has long been central to research in social mobility, demography, migration, and epidemiology, yet it remains challenging due to name variation, missing or inconsistent attributes, and changes in administrative boundaries over time [23, 5, 17]. Prior work on census linkage has largely relied on proprietary or restricted-access data, often focusing on short temporal windows or single national censuses. While large-scale projects such as IPUMS provide extensive longitudinal census coverage, redistribution constraints and limited availability of downloadable record-level data restrict their use as open benchmarks [25, 24]. Publicly available datasets that combine long temporal horizons with genealogical structure and explicit identity labels remain rare and expensive to produce.

Classical entity resolution benchmarks. Most widely used entity resolution (ER) benchmarks were developed for product matching or bibliographic deduplication tasks, including Abt-Buy, Amazon-Google, DBLP-ACM, and related datasets [14, 10]. These benchmarks are typically static snapshots, contain flat textual attributes, and lack temporal structure, demographic features, or relational context. As noted by Papadakis *et al.* [21], many such datasets are comparatively easy, with high performance achievable via simple similarity thresholds [21, 11, 20]. As a result, they fail to capture the ambiguity, non-stationarity, and structural complexity characteristic of real-world population data.

Longitudinal and temporally structured benchmarks. Several recent benchmarks emphasize temporal or domain shift in tabular data, including TableShift [6] and TabReD [22]. These datasets demonstrate how distributional change can substantially affect model performance, but they do not include identity resolution tasks, genealogical relationships, or hierarchical geographic structure. Other longitudinal resources track identities over time in different domains—for example ORCID (researcher identities), parliamentary corpora such as ParlaMint, and correspondence corpora such as correspSearch—but they target different entity types and generally lack census-style demographic density, household structure, or kinship links [19, 18, 4, 15].

Positioning of ICE-ID. ICE-ID complements existing work by providing an open benchmark for person-level identity resolution under long-term temporal drift. Unlike classical ER datasets, which are typically static and attribute-light, ICE-ID captures non-stationarity, name collisions, and structured geographic context in historical records. Unlike many longitudinal resources, which are restricted-access or centered on other entity types, ICE-ID is released with fixed temporal evaluation protocols and reproducible analysis artifacts. It therefore fills a gap between static ER benchmarks and large but non-redistributable census collections, enabling reproducible research on longitudinal identity resolution in realistic settings [9].

3 Dataset

ICE-ID is a longitudinal census benchmark for person-level identity resolution under historical temporal drift. It contains **984,028 records** from **16 Icelandic census waves** spanning **1703–1920**, with **226,864** expert-curated person identifiers. Each row corresponds to one individual appearance in one census wave, allowing repeated observations across time.

The dataset captures several sources of real-world complexity that are largely absent from standard entity resolution benchmarks. Personal names follow Icelandic patronymic conventions, leading to systematic name collisions across the population. Demographic attributes such as birth year and marital status exhibit natural reporting noise and temporal inconsistency. Geographic information is encoded hierarchically (farm, parish, district, county) and evolves over time due to administrative boundary changes. Kinship links (partner, father, mother) are available but sparsely populated, reflecting historical census recording practices.

The release packages census appearances and expert-curated identity labels together with geographic tables encoding Iceland's evolving administrative hierarchy. Ground-truth clusters are defined by the stable `person` identifier, enabling both pairwise linkage and multi-record clustering evaluation within and across census waves. This structure supports reproducible longitudinal ER experiments without relying on restricted-access census collections.

3.1 Schema and Tables

ICE-ID comprises four CSV files organized into two components:

1. **Geographic tables** (`counties.csv`, `districts.csv`, `parishes.csv`): Encode Iceland's evolving territorial hierarchy. Each record carries a unique identifier, human-readable name, validity interval ("begins"/"ends"), and geographic centroids (`lat`, `lon`).
2. **People table** (`people.csv`): 984,028 rows—one per individual appearance in each census wave (1703–1920). Each row captures:
   - Name components: `nafn_norm`, `first_name`, `patronym`, `surname`
   - Demographics: `birthyear`, `sex`, `status`, `marriagestatus`
   - Cluster labels: `person` (expert-curated identity)
   - Kinship links: `partner`, `father`, `mother` with provenance tags
   - Geography: `farm`, `parish`, `district`, `county`

4 Temporal Coverage and Label Density

Figure 1 shows the distribution of records across census waves and the proportion with cluster labels.

Key observations: ICE-ID spans 16 census waves from 1703 to 1920. Record counts range from 8,072 (1729, partial) to 102,699 (1920). The total labeled population comprises 226,864 unique person identifiers, with 106,168 persons (46.8%) appearing in multiple waves. The average label rate is 50.17%.

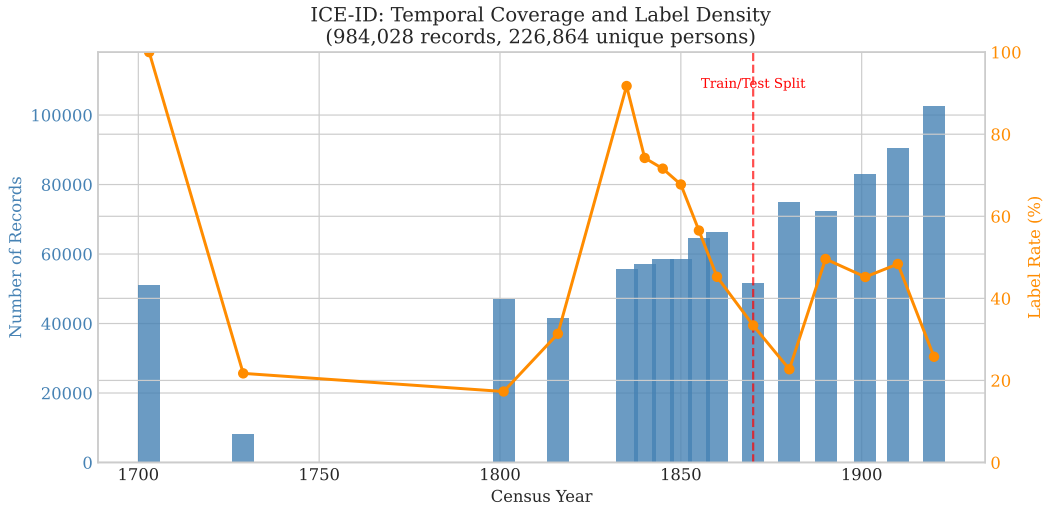

Figure 1: Temporal coverage and label density. ICE-ID spans 16 census waves; the 1703 census contains 50,959 records and the 1920 census contains 102,699. The average label rate (records with `person` assigned) is 50.17%. Classic ER datasets are single-snapshot ("static") and their label density is computed as the fraction of records appearing in at least one positive match pair.

5 Missingness Over Time by Feature Family

Understanding missing data patterns is an important step prior to model development [16]. Figure 2 shows missing-rate trajectories over time for four feature families. All numeric values below are taken from `bench/paper_artifacts/plot_data/fig2_missingness.json`.

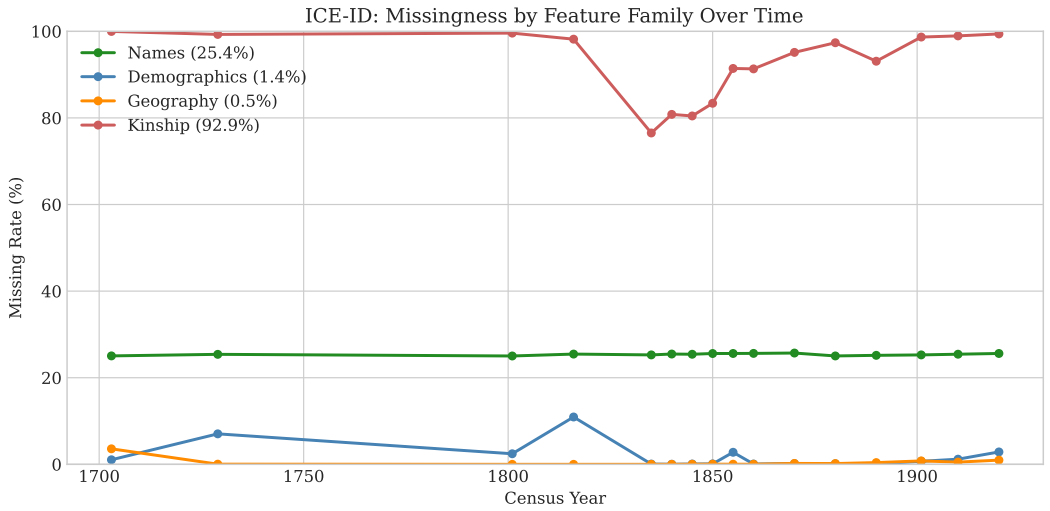

Figure 2: Missingness rates by feature family across census waves. Names are 25.39% missing overall (dominated by surname), demographics are 1.38% missing overall, geography is near-complete, and kinship links are 92.91% missing overall.

Key observations: Names are 25.39% missing overall, driven primarily by surname. Demographics are 1.38% missing overall (with most waves under 3%). Kinship links are 92.91% missing overall, explaining why methods that rely on household structure must be robust to sparse relational evidence.

6 Entity Cluster Size Distribution

The cluster size distribution—how many census appearances per person—directly affects entity resolution difficulty. Figure 3 shows the log–log CCDF.

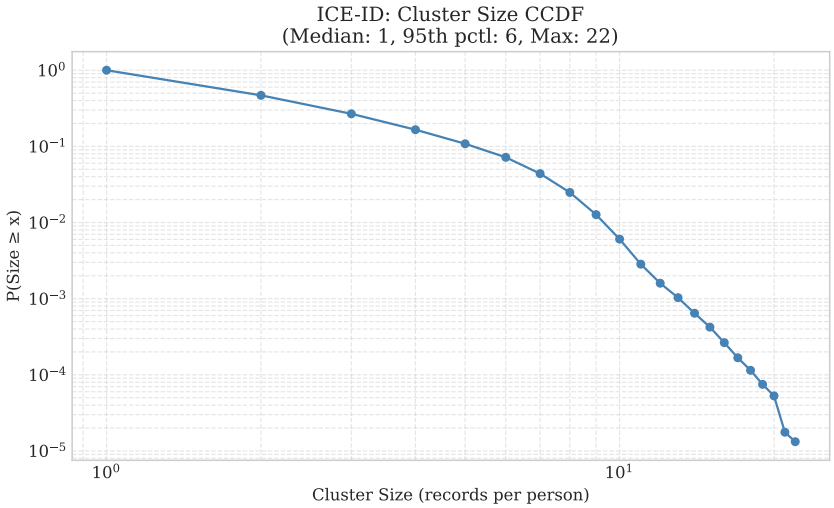

Figure 3: Cluster size CCDF (log–log). Median cluster size is 1; 95th percentile is 6; maximum cluster size is 22.

Key observations: 120,696 persons (53.2%) appear in only one census; 45,436 (20.0%) appear in two. The maximum cluster size is 22 appearances. Median cluster size is 1; 95th percentile is 6. The long tail of large clusters (individuals appearing in 6+ censuses) represents high-value longitudinal subjects but also presents challenges for clustering algorithms that must maintain transitivity over many pairwise predictions.

7 Identifier Ambiguity: Name Collision Analysis

Patronymic naming conventions in Iceland create significant name collisions. Figure 4 shows the Zipf distribution of normalized names and compares token entropy to classic ER datasets.

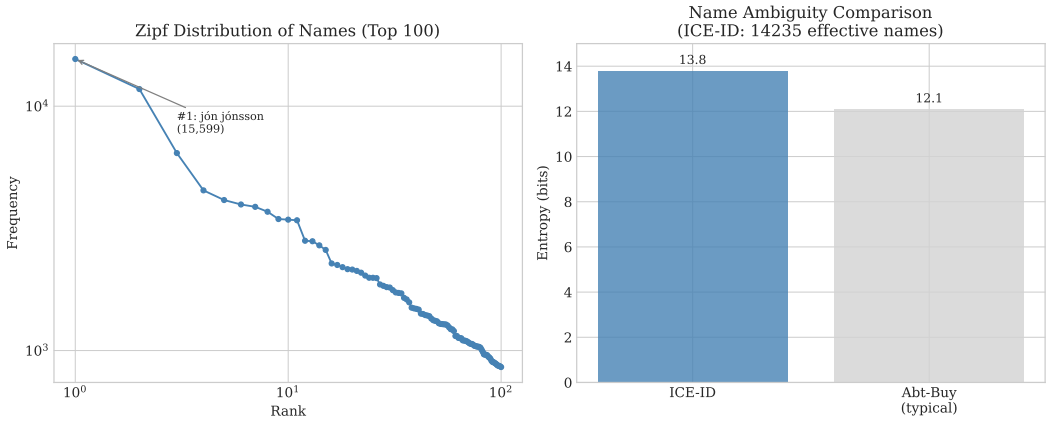

Figure 4: Name ambiguity analysis. (Left) Zipf plot of top 100 normalized names (`nafn_norm`). (Right) Token entropy comparison between ICE-ID and representative classic ER datasets.

Key observations: The most common name ("Jón Jónsson") appears 15,599 times. The top 10 names account for 6.2% of all records. While the entropy (13.8 bits) is comparable to product datasets, the

patronymic naming system creates systematic collisions: many individuals share identical names. This motivates the use of additional signals (birthyear, geography, kinship) for accurate disambiguation.

8 Candidate Generation Efficiency

Scalable entity resolution requires effective blocking to reduce the $O(n^2)$ comparison space [3, 12]. Figure 5 shows blocking recall vs. candidate budget for different strategies.

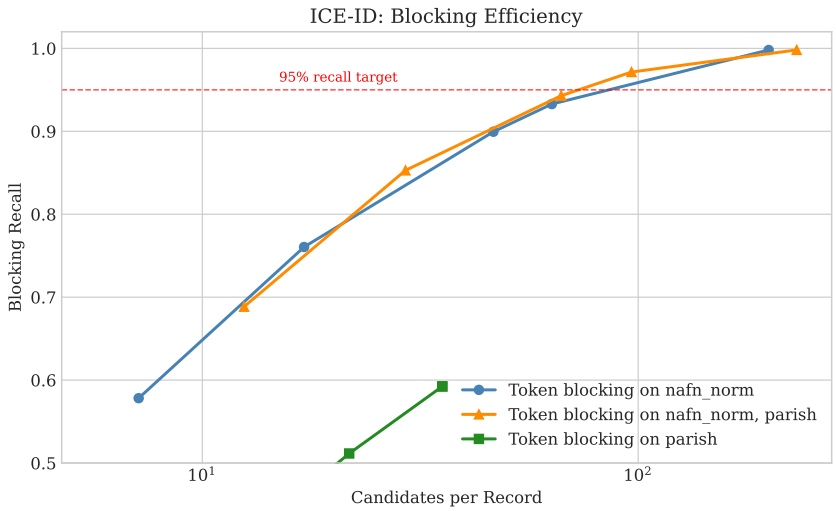

Figure 5: Blocking efficiency curves. Token blocking on `nafn_norm` achieves 0.90 recall at 46.5 candidates/record; at 199 candidates/record it reaches 0.998 recall. Hybrid token blocking (name+parish) achieves 0.94 recall at 66.6 candidates/record and 0.97 recall at 96.5 candidates/record.

Key observations: Pure geographic blocking (parish only) is insufficient due to population mobility and administrative boundary changes, achieving only 0.59 recall even with 35 candidates/record. Name-based blocking on `nafn_norm` is more effective, reaching 0.90 recall at 46 candidates/record. The hybrid strategy (name + parish) provides the best recall/efficiency tradeoff, achieving 0.97 recall at 96.5 candidates/record; on this 7,466-record subset this corresponds to about 2.59% of all possible unordered pairs.

9 Comparison with Classical ER Benchmarks

Table 1 compares ICE-ID against standard entity resolution datasets.

Table 1: Dataset synopsis: ICE-ID vs. classical ER benchmarks.

Dataset	Time Span	#Waves	#Records	#Labeled IDs	%Labeled	Geo	Kinship
ICE-ID	1703–1920	16	984,028	226,864	50.2%	Hierarchical	Yes
Abt-Buy	N/A	1	11,486	11,486	100%	Flat	No
Amazon-Google	N/A	1	13,748	13,748	100%	Flat	No
DBLP-ACM	N/A	1	14,834	14,834	100%	Flat	No
DBLP-Scholar	N/A	1	34,446	34,446	100%	Flat	No
Walmart-Amazon	N/A	1	12,288	12,288	100%	Flat	No
iTunes-Amazon	N/A	1	642	642	100%	Flat	No
Beer	N/A	1	536	536	100%	Flat	No
Fodors-Zagats	N/A	1	1,134	1,134	100%	Flat	No

Table 2 provides a schema comparability matrix showing feature availability across datasets.

Key observations: ICE-ID is the only dataset combining temporal coverage, hierarchical geography, and kinship signals. Classical ER benchmarks lack demographic attributes entirely and provide only flat text fields for matching. This makes ICE-ID uniquely suited for evaluating methods that leverage structured signals.

Table 2: Schema comparability matrix. ✓= present, ∼ = partial, — = absent.

Feature Family	ICE-ID	Abt-Buy	Amz-Ggl	DBLP-ACM	Wlm-Amz	iTun-Amz	Beer	Fod-Zag
Name / Title	✓	✓	✓	✓	✓	✓	✓	✓
Age / Birthyear	✓	—	—	—	—	—	—	—
Sex / Gender	✓	—	—	—	—	—	—	—
Household / Family	✓	—	—	—	—	—	—	—
Parent links	✓	—	—	—	—	—	—	—
Spouse / Partner	✓	—	—	—	—	—	—	—
Address / Geo	✓(4-level)	—	—	—	—	—	—	✓
Temporal field	✓	—	—	∼ (year)	—	∼ (released)	—	—
Free-text notes	∼	✓	✓	—	✓	—	—	—

10 Comparison with Longitudinal Datasets

While classical ER benchmarks are static snapshots, several other datasets capture identity over time. Table 3 compares ICE-ID to longitudinal identity datasets.

Table 3: Longitudinal dataset comparison. "File" = downloadable data; "Doc" = metadata only.

Dataset	Time Span	Entity Type	∼Entities	Temporal Signal	Data	Access
ICE-ID	1703–1920	Person	227K	Census year	File	Open
IPUMS LRS	1850–1940	Person	50M	Census year	Doc	Account
IPUMS MLP	1870–2020	Person	100M	Census+survey	Doc	Account
IPUMS NAPP	1801–1910	Person	100M	Census year	Doc	Account
ORCID	2012–present	Researcher	18M	Last modified	File (sample)	Open
SemParl	1907–2021	Parliamentarian	7K	Speech date	File	Open
CKCC	1600–1800	Correspondent	5K	Letter date	File	Open
correspSearch	1500–2000	Correspondent	130K letters	Letter date	File	Open
Synthea	1950–2020	Patient	1K (sample)	Encounter date	File	Open
FEBRL	N/A	Patient	5–10K	None (static)	File	Open

Key observations: IPUMS datasets provide the largest historical census coverage but require account access and do not function as fully redistributable ER benchmarks. Open alternatives (ORCID, parliamentary corpora, correspondence KGs) cover different entity types and temporal signals. ICE-ID combines a multi-century census setting with hierarchical geography and kinship fields in an openly downloadable benchmark release.

11 Missing Data Pattern Visualization

Figure 6 provides a visual summary of missing data patterns in ICE-ID using a matrix plot. Each column represents a feature; white cells indicate missing values.

Key observations: The visualization confirms quantitative findings from Section 5: geographic hierarchy is reliably present, while kinship links are sparsely populated. This visual pattern is characteristic of historical census data where household relationships were inconsistently recorded.

12 Benchmark Tasks and Protocols

ICE-ID supports evaluation of identity resolution methods under realistic temporal distribution shift. We define two canonical tasks and a strictly temporal evaluation protocol intended to reflect deployment settings where early census waves are available for training while later waves are held out for testing.

12.1 Tasks

We define two benchmark tasks:

1. **Within-wave (intra-census) linkage:** determine whether two records from the same census wave refer to the same individual.

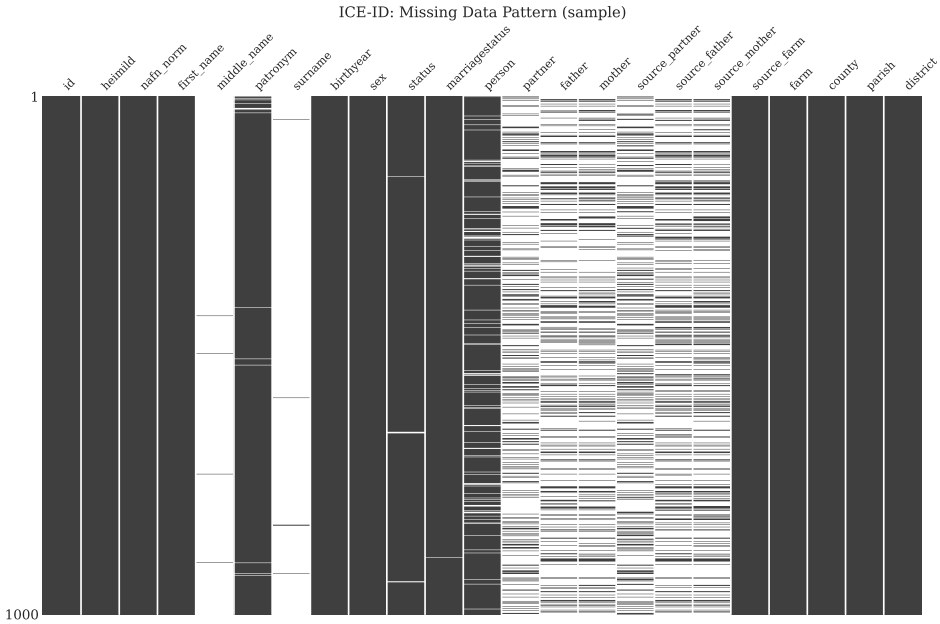

Figure 6: Missing data matrix (sample of 1,000 records). Geographic fields are near-complete; kinship links (partner, father, mother) are mostly missing; names show moderate missingness.

2. **Across-wave (cross-census) linkage:** determine whether two records from different census waves refer to the same individual, despite multi-decadal drift in name spelling, reported birthyear, residence, and administrative boundaries.

Unless otherwise specified, cross-census evaluation is performed on records drawn from adjacent census waves within a split; the protocol can also be applied to non-adjacent waves to probe longer-range temporal drift.

12.2 Temporal splits

We use a deployment-faithful temporal split:

- **Train:** pre-1870 (560,334 records; 153,311 unique persons)
- **Validation:** 1871–1890 (147,450 records; 44,645 unique persons)
- **Test:** 1891–1920 (276,244 records; 62,109 unique persons)

This split induces a temporal out-of-distribution (OOD) setting in which the model must generalize from earlier censuses to later ones. A small cross-split overlap exists: 6,136 persons (2.7% of unique persons) appear in both train and test splits, enabling direct evaluation of temporal generalization on individuals observed across long time spans.

12.3 Ground truth and pair construction

Ground-truth identities are defined by the expert-curated cluster label `person`. A **positive pair** is any pair of records within the evaluation split that share the same `person` identifier. **Negative pairs** are sampled from records with different `person` identifiers. To limit trivial negatives and reflect realistic candidate sets, negatives are sampled *within the same blocking partition* (e.g., as produced by a candidate-generation strategy), using a default ratio of two negatives per positive [27].

12.4 Evaluation modes and metrics

ICE-ID supports evaluation in both **pairwise** and **clustering** modes.

- **Pairwise evaluation** scores predicted match decisions on labeled pairs using standard metrics such as precision, recall, and F_1.
- **Clustering evaluation** assesses the quality of predicted identity clusters against ground-truth `person` clusters using clustering metrics such as Adjusted Rand Index (ARI).

For clustering-based systems, outputs should respect **transitivity**: if record A matches B and B matches C, then A and C should be placed in the same predicted entity cluster. Reporting both pairwise and clustering metrics is recommended, as high pairwise accuracy does not necessarily imply globally coherent clusters under transitivity constraints.

12.5 Protocol Summary

Table 4 summarizes the canonical evaluation protocol for ICE-ID.

Table 4: Evaluation protocols and temporal splits.

Component	Specification
Temporal splits	Train: up to 1870 (560,334 records, 153,311 unique persons); Val: 1871–1890 (147,450 records, 44,645 persons); Test: 1891–1920 (276,244 records, 62,109 persons)
Positive pairs	All pairs of records sharing the same `person` ID within the evaluation split
Negative sampling	2 negatives per positive, sampled from same blocking partition
Transitivity	Ground-truth clusters defined by `person` field; methods should enforce transitivity in clustering outputs
Evaluation modes	Within-wave (same census) and cross-wave (adjacent censuses)

13 Dataset Card

Following the Datasheets for Datasets framework [7, 2, 13]:

13.1 Motivation and Intended Use

Purpose: Provide a realistic, large-scale benchmark for longitudinal identity resolution.

Intended Uses: (1) Entity resolution research; (2) Temporal distribution shift studies; (3) Genealogical/historical research.

Out-of-Scope Uses: (1) Re-identification of living individuals (data ends 1920); (2) Commercial genealogy without attribution.

13.2 Composition

Records: 984,028 individual census appearances. **Labeled Entities:** 226,864 unique person clusters (106,168 with multiple records). **Fields:** 23 columns including names, demographics, kinship, geography.

13.3 Collection and Labeling

Collection: Digitized from Icelandic census manuscripts by National Archives. **Labels:** Expert-curated using genealogical records and parish registers.

13.4 Maintenance

License: CC-BY-4.0.
Dataset Access: `https://huggingface.co/datasets/goldpotatoes/ice-id`
Code and Artifacts: `https://github.com/IIIM-IS/ICE-ID-2.0`

13.5 Interactive Exploration and Model Tooling

To facilitate engagement with the ICE-ID ecosystem, we provide an interactive Streamlit-based dashboard for end-to-end entity resolution and data exploration (see Figure 7). The application is organized into four primary workspaces:

v2/figures/dashboard1.png

(a) Cluster browsing interface showing edge navigation, prediction-vs-ground-truth comparison, and colour-coded member details for a selected cluster.

v2/figures/dashboard2.png

(b) Interactive match network of predicted edges (left) and genealogical tree of family relationships (right) for visual validation of linked records.

Figure 7: The ICE-ID Dashboard: (a) interactive visualization of identity clusters and (b) diagnostic suite for model performance and missingness analysis.

- **Training:** Supports the configuration and execution of the ICE-ID pipeline alongside 12 external baselines (e.g., *Ditto*, *XGBoost*, *Splink*) with real-time log streaming and hyperparameter management.
- **Evaluation:** Computes pairwise metrics with Wilson confidence intervals and cluster-level statistics (weighted purity, singleton rates), generating automated interpretive commentary for results.
- **Inspect & Edit:** A debugging interface featuring `pyvis`-based match networks and genealogical trees. It allows for live edge manipulation with automatic Union-Find cluster recomputation and a feedback loop for model finetuning via JSON-based label exports.
- **Data Editor:** Provides tools for reshaping the underlying CSVs, visualizing missing-data patterns (matrix/correlation plots), and managing persistent model configuration profiles.

14 Limitations and Ethical Considerations

Limitations:

- **Snapshot waves only:** Continuous life-course trajectories are inferred, not observed.
- **Kinship sparsity:** Only 7% of records have partner/parent links.
- **Label noise:** Early censuses (1703, 1729) have higher transcription error rates.
- **Coverage gaps:** 1729 and 1870 censuses are partial.

Ethical considerations: Two facts minimize ethical issues: firstly, the family-link records are sparsely populated such that many persons lack explicit relational connections to other persons, for example, parental relationship or marital counterparts. Secondly, census data ends in 1920, so no direct link to living individuals exists. The dataset enables historical/demographic research while minimizing re-identification risks.

15 Conclusion

ICE-ID provides a unique resource for entity resolution research: a century-spanning, genealogically enriched dataset with hierarchical geography and temporal drift. Its structure exposes challenges (name ambiguity, missingness, cluster size heterogeneity) absent from classical benchmarks. By releasing data, protocols, and analysis artifacts, we enable reproducible research on longitudinal identity resolution. For model evaluations and method comparisons, see our companion paper.

Acknowledgements

We acknowledge the **Center of Digital Humanities and Arts (CDHA), Iceland**, for enabling access to and supporting the use of the historical census and registry data underlying ICE-ID. We thank the technical and administrative staff involved in the digitization, preservation, and curation of these records, whose work made this dataset available for research. We also acknowledge institutional support from the participating universities and research centers that facilitated data processing, analysis, and the open release of this benchmark.

References

[1] Martha J. Bailey, Connor Cole, Morgan Henderson, and Catherine Massey. How well do automated linking methods perform? lessons from us historical data. *Journal of Economic Literature*, 58(4):997–1044, December 2020. doi: 10.1257/jel.20191526. URL https://www.aeaweb.org/articles?id=10.1257/jel.20191526.

[2] Emily M. Bender and Batya Friedman. Data statements for natural language processing: Toward mitigating system bias and enabling better science. *Transactions of the Association for Computational Linguistics*, 6:587–604, 2018. URL https://api.semanticscholar.org/CorpusID:52255687.

[3] Peter Christen. A survey of indexing techniques for scalable record linkage and deduplication. *IEEE Transactions on Knowledge and Data Engineering*, 24:1537–1555, 2012. URL https://api.semanticscholar.org/CorpusID:2082499.

[4] Tomaž Erjavec et al. The parlamint corpora of parliamentary proceedings. *Language Resources and Evaluation*, 57(4):1265–1324, 2022. doi: 10.1007/s10579-022-09637-0. URL https://doi.org/10.1007/s10579-022-09637-0.

[5] Ivan P. Fellegi and Alan B. Sunter. A theory for record linkage. *Journal of the American Statistical Association*, 64(328):1183–1210, 1969. doi: 10.1080/01621459.1969.10501049. URL https://www.tandfonline.com/doi/abs/10.1080/01621459.1969.10501049.

[6] Josh Gardner, Zoran Popovic, and Ludwig Schmidt. Benchmarking distribution shift in tabular data with tableshift. *ArXiv*, abs/2312.07577, 2023. URL https://api.semanticscholar.org/CorpusID:264546341.

[7] Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé III, and Kate Crawford. Datasheets for datasets. *Communications of the ACM*, 64(12):86–92, 2021.

[8] Hannah Rose Kirk, Alexander Whitefield, Paul Rottger, Andrew M. Bean, Katerina Margatina, Juan Ciro, Rafael Mosquera, Max Bartolo, Adina Williams, He He, Bertie Vidgen, and Scott A. Hale. The prism alignment dataset: What participatory, representative and individualised human feedback reveals about the subjective and multicultural alignment of large language models. In *Neural Information Processing Systems*, 2024. URL https://api.semanticscholar.org/CorpusID:269362843.

[9] Pradap Konda, Jeff Naughton, Shishir Prasad, Ganesh Krishnan, Rohit Deep, Vijay Raghavendra, Sanjib Das, Paul C., AnHai Doan, Adel Ardalan, Jeffrey Ballard, Han Li, Fatemah Panahi, and Haojun Zhang. Magellan: toward building entity matching management systems over data science stacks. *Proceedings of the VLDB Endowment*, 9:1581–1584, 09 2016. doi: 10.14778/3007263.3007314.

[10] Hanna Köpcke, Andreas Thor, and Erhard Rahm. Evaluation of entity resolution approaches on real-world match problems. *Proceedings of the VLDB Endowment*, 3:484 – 493, 2010. URL https://api.semanticscholar.org/CorpusID:15245020.

[11] Yuliang Li, Jinfeng Li, Yoshihiko Suhara, AnHai Doan, and Wang Chiew Tan. Deep entity matching with pre-trained language models. *Proceedings of the VLDB Endowment*, 14:50 – 60, 2020. URL https://api.semanticscholar.org/CorpusID:214743579.

[12] Matthew Michelson and Craig A. Knoblock. Learning blocking schemes for record linkage. In *AAAI Conference on Artificial Intelligence*, 2006. URL https://api.semanticscholar.org/CorpusID:6199729.

[13] Margaret Mitchell, Simone Wu, Andrew Zaldivar, Parker Barnes, Lucy Vasserman, Ben Hutchinson, Elena Spitzer, Inioluwa Deborah Raji, and Timnit Gebru. Model cards for model reporting. *Proceedings of the Conference on Fairness, Accountability, and Transparency*, 2018. URL https://api.semanticscholar.org/CorpusID:52946140.

[14] Sidharth Mudgal, Han Li, Theodoros Rekatsinas, AnHai Doan, Youngchoon Park, Ganesh Krishnan, Rohit Deep, Esteban Arcaute, and Vijay Raghavendra. Deep learning for entity matching: A design space exploration. 2018. URL https://api.semanticscholar.org/CorpusID:44063437.

[15] Martin Anton Müller and Stefan Dumont. <correspdesc> connecting scholarly editions of letters. *Journal of the Text Encoding Initiative*, (10), 2017. doi: 10.4000/jtei.1742. URL https://doi.org/10.4000/jtei.1742.

[16] Felix Naumann. Data profiling revisited. *SIGMOD Rec.*, 42:40–49, 2014. URL https://api.semanticscholar.org/CorpusID:7801494.

[17] Howard B. Newcombe, James M. Kennedy, Samantha Axford, and Allen P. James. Automatic linkage of vital records. *Science*, 130 3381:954–9, 1959. URL https://api.semanticscholar.org/CorpusID:1471056.

[18] ORCID. Annual public data files. https://info.orcid.org/annual-public-data-file-on-orcid-records/. Accessed: 2026-02-23.

[19] ORCID. About orcid. https://orcid.org/about, 2024. Accessed: 2026-02-23.

[20] George Papadakis, Dimitrios Skoutas, Emmanouil Thanos, and Themis Palpanas. A survey of blocking and filtering techniques for entity resolution. *ArXiv*, abs/1905.06167, 2019. URL https://api.semanticscholar.org/CorpusID:155089485.

[21] George Papadakis, Nishadi Kirielle, Peter Christen, and Themis Palpanas. A critical re-evaluation of benchmark datasets for (deep) learning-based matching algorithms. *arXiv preprint arXiv:2307.01231*, 2023. URL https://arxiv.org/abs/2307.01231.

[22] Ivan Rubachev, Nikolay Kartashev, Yury Gorishniy, and Artem Babenko. Tabred: Analyzing pitfalls and filling the gaps in tabular deep learning benchmarks, 2024. URL https://arxiv.org/abs/2406.19380.

[23] Steven Ruggles, Catherine A. Fitch, and Evan Roberts. Historical census record linkage. *Annual review of sociology*, 44:19–37, 2018. URL https://api.semanticscholar.org/CorpusID:53059180.

[24] Adrian Sayers, Yoav Ben-Shlomo, Ashley W. Blom, and Fiona Steele. Probabilistic record linkage. *International Journal of Epidemiology*, 45:954 – 964, 2015. URL https://api.semanticscholar.org/CorpusID:15748808.

[25] Matthew Sobek, Steven Ruggles, Sarah Menser, and Others. Big data: Large-scale historical infrastructure from the minnesota population center. *Historical Methods: A Journal of Quantitative and Interdisciplinary History*, 44(2):61–68, 2011.

[26] Boxin Wang, Weixin Chen, Hengzhi Pei, Chulin Xie, Mintong Kang, Chenhui Zhang, Chejian Xu, Zidi Xiong, Ritik Dutta, Rylan Schaeffer, Sang T. Truong, Simran Arora, Mantas Mazeika, Dan Hendrycks, Zinan Lin, Yu Cheng, Sanmi Koyejo, Dawn Song, and Bo Li. Decodingtrust: A comprehensive assessment of trustworthiness in gpt models, 2024. URL https://arxiv.org/abs/2306.11698.

[27] Steven Whang, David Marmaros, and Hector Garcia-Molina. Pay-as-you-go entity resolution. *Knowledge and Data Engineering, IEEE Transactions on*, 25:1111–1124, 05 2013. doi: 10.1109/TKDE.2012.43.

[28] Sungduk Yu, Zeyuan Hu, Akshay Subramaniam, Walter Hannah, Liran Peng, Jerry Lin, Mohamed Aziz Bhouri, Ritwik Gupta, Björn Lütjens, Justus C. Will, Gunnar Behrens, Julius J. M. Busecke, Nora Loose, Charles I. Stern, Tom Beucler, Bryce Harrop, Helge Heuer, Benjamin R. Hillman, Andrea Jenney, Nana Liu, Alistair White, Tian Zheng, Zhiming Kuang, Fiaz Ahmed, Elizabeth Barnes, Noah D. Brenowitz, Christopher Bretherton, Veronika Eyring, Savannah Ferretti, Nicholas Lutsko, Pierre Gentine, Stephan Mandt, J. David Neelin, Rose Yu, Laure Zanna, Nathan Urban, Janni Yuval, Ryan Abernathey, Pierre Baldi, Wayne Chuang, Yu Huang, Fernando Iglesias-Suarez, Sanket Jantre, Po-Lun Ma, Sara Shamekh, Guang Zhang, and Michael Pritchard. Climsim-online: A large multi-scale dataset and framework for hybrid ml-physics climate emulation, 2024. URL https://arxiv.org/abs/2306.08754.

A Dataset Access

Full dataset available at https://huggingface.co/datasets/goldpotatoes/ice-id.

B Supplementary Statistics

Detailed per-census statistics and additional analysis artifacts are provided in the repository.

C Artifact Provenance Graph

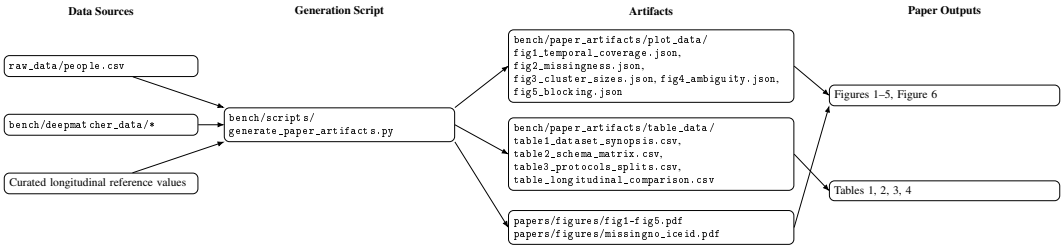

Figure 8: Data-paper provenance graph mapping source data, generation script, saved artifacts, and paper figures/tables.